



UiT The Arctic University of Norway

Machine Learning and Industrial Applications

Industry X.0 - INE-3612

Wahyu Andy Prastyabudi

Department of Industrial Engineering
Faculty of Engineering Science and Technology

2026-08-26



Introduction

Key Concepts of ML

Algorithms and Techniques

Applications of ML

Final Thoughts

Introduction

AI Layers

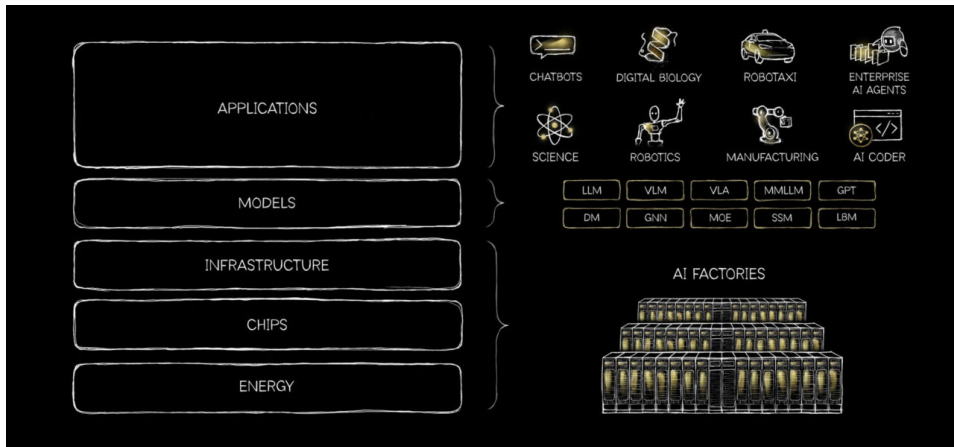


Figure 1: AI layers defined by Jensen Huang - NVIDIA. Image is taken from nvidia.com

What is machine learning (ML)?

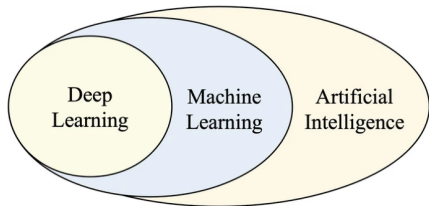


Figure 2: Subfields of AI. Taken from Wang (2025)

- ML is a subfield of AI focusing on building a machine (i.e., model) to learn from experience with minimal human intervention
- ML finds generalizable predictive patterns from the data
- It involves mathematical models for approximating the data

What is machine learning (ML)? (cont.)

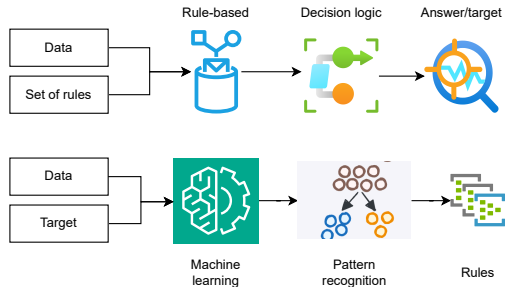


Figure 3: Rule-based learning vs machine learning

- Rule-based systems require pre-existing knowledge to build robust set of rules
- Rule-based is often seen in expert systems, aiming at modeling human domain experts to solve specific tasks
- ML allows to detect the pattern automatically using the data without explicit rule-based instruction
- It uses the rules to predict future unseen data
- ML has capabilities to extract knowledge from data and improve the performance (i.e., adaptive learning capability)

Why ML matters

Future of Jobs Report 2025

Trends in AI, robotics and energy technologies are most likely to drive business transformation



Share of employers expecting the technology to transform their organization

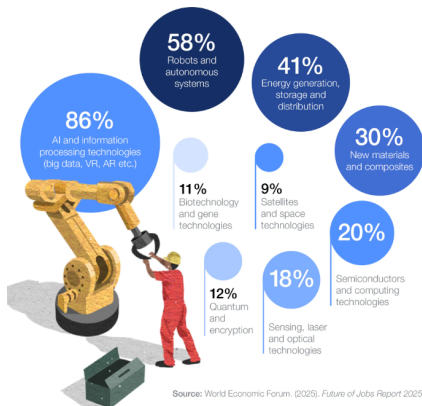


Figure 4: Taken from <https://www.weforum.org>

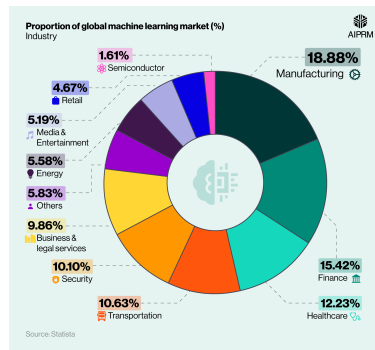


Figure 5: Taken from <https://www.aiprm.com>

“Machine learning is changing, or will change, every industry, and leaders need to understand the basic principles, the potential, and the limitations.”

Aleksander Madry - MIT Computer Science Professor

Key Concepts of ML

- **Data-driven**
 - ▶ The goal is to extract valuable patterns from data
 - ▶ Data quality plays a critical role in producing reasonable outputs
- **Model**
 - ▶ It maps the input features to target outputs
 - ▶ Good models generalize well to unseen data
- **Learning**
 - ▶ Automatic pattern recognition
 - ▶ Parameters optimization of the model

Types of ML

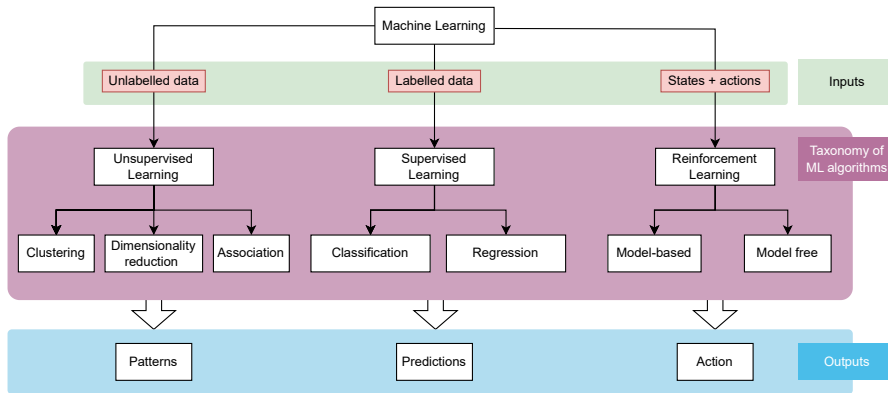


Figure 6: Taxonomy of ML algorithms

What's behind ML?

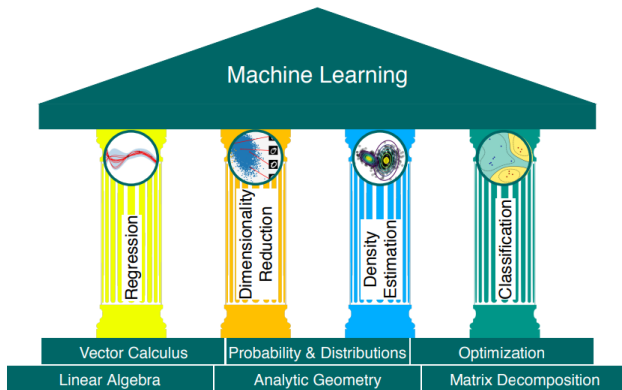


Figure 7: Foundation of ML. Taken from Deisenroth et al. (2020)

Further reading: Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). *Mathematics for machine learning*. Cambridge University Press.

General Workflow

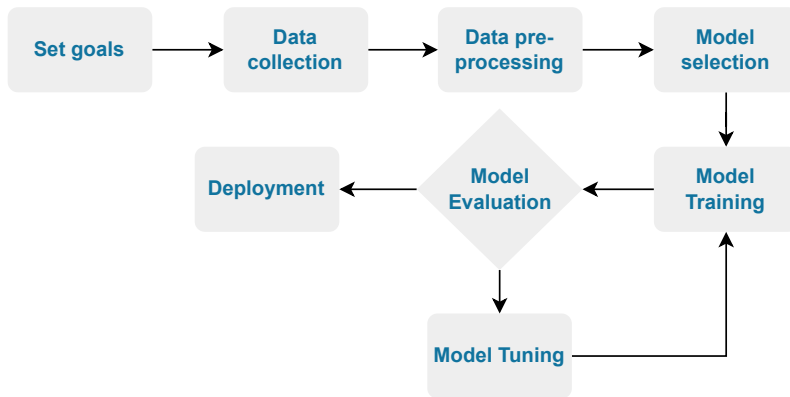


Figure 8: Typical ML workflow

Algorithms and Techniques



Figure 9: Unsupervised learning illustration

Table 1: Unsupervised learning algorithms.

Category	Algorithm	Use case example
Clustering	K-means	Customer segmentation, supplier performance evaluation
	Dbscan	Detecting anomalies in delivery patterns
	Hierarchical	Analysing production batch similarities
Dimension reduction	PCA	Visualizing high-dimensional sensor data
Association mining	Apriori	Discover frequent items in order transaction
	FP-growth	Identifying co-occurring parts in inventory

Popular Algorithms (cont.)

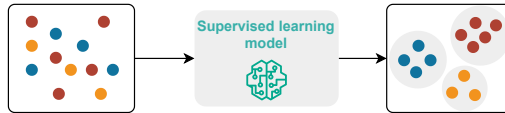


Figure 10: Supervised learning illustration

Table 2: Supervised learning algorithms.

Category	Algorithm	Use case example
Regression	Linear regression	Predicting maintenance cost
	Logistic regression	Predicting machine failure
Classification	KNN	Classifying defective and non-defective products
	SVM	Safety incidents classification
	Naïve Bayes	Transaction records filtering
	Neural Network (NN)	Product quality monitoring using image classification

Popular Algorithms (cont.)

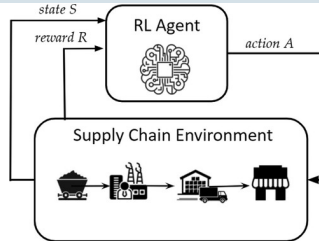


Figure 11: Reinforcement learning in SCM. Taken from Rolf et al. (2023)

Table 3: Reinforcement learning algorithms.

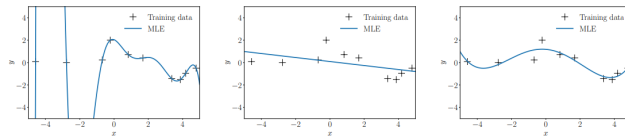
Category	Algorithm	Use case example
Model-free	Q-learning	Optimizing robot navigation, optimizing inventory
	Monte Carlo methods	Evaluating long-term supply chain policies
Model-based	Markov decision process (MDP)	Dynamic scheduling in manufacturing lines

- Validate the model against unseen dataset
- Standardize model's performance
- Measure model's ability to generalize the data
- Identify strategies (e.g., pre-processing, cross-validation, sampling) for improvement
- Measure choice depends on the nature of problem

Table 4: ML evaluation metrics.

ML Type	Metric	What it measures
Supervised (classification)	Accuracy	Percentage of correct prediction
	Precision	Reliability of positive prediction
	Recall	Ability to find all true cases
Supervised (regression)	MAE	Average of errors (i.e., gap between estimate and true value)
	RMSE	Square root of the average squared errors
Unsupervised	Silhouette Coefficient	Measure the similarity of a sample point in its own cluster vs others
	Davies-Bouldin Index	Measure average similarity between clusters
Reinforcement	Cumulative reward	Total rewards or points over time
	Episode length	Steps taken to complete the task

Model Fitting



(a) Overfitting

(b) Underfitting.

(c) Fitting well.

Figure 12: Fitting problems in ML. Taken from Deisenroth et al. (2020).

Model Fitting

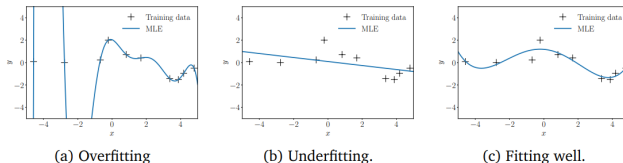


Figure 12: Fitting problems in ML. Taken from Deisenroth et al. (2020).

Underfitting

- Build more complex model
- Add relevant features
- Feature engineering
- Check data quality
- Adjust training parameters (e.g., reduce regularization)

Overfitting

- Perform appropriate cross-validation
- Apply regularization
- Remove irrelevant features
- Simplify the model
- Consider early stopping

Applications of ML

ML in real-world applications

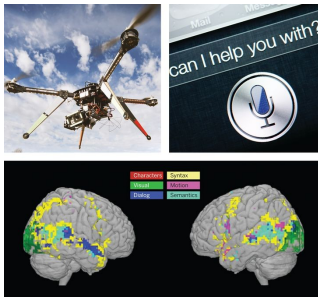


Figure 13: Robotics and autonomous control, speech recognition, neuroscience. Taken from Jordan and Mitchell (2015)

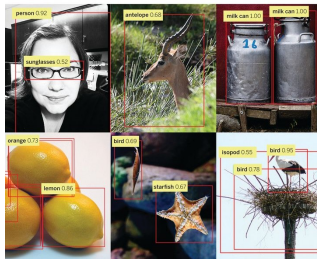


Figure 14: Object detection. Taken from Jordan and Mitchell (2015)

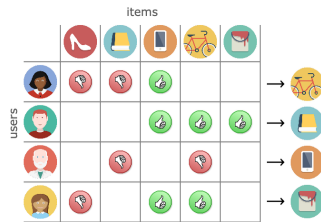


Figure 15: Recommender systems. Taken from www.wolfram.com

Supply Chain Case

Cavalcante, I. M., Frazzon, E. M., Forcellini, F. A. & Ivanov, D. (2019). A supervised machine learning approach to data-driven simulation of resilient supplier selection in digital manufacturing. *International Journal of Information Management*, 49, 86–97.

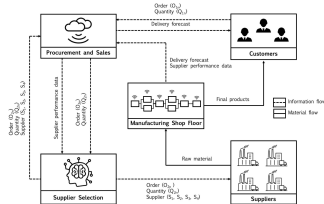


Figure 16: Make to order simulation model

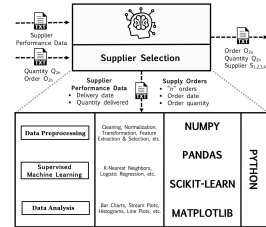


Figure 17: Supplier selection with supervised learning

Cavalcante, I. M., Frazzon, E. M., Forcellini, F. A. & Ivanov, D. (2019). A supervised machine learning approach to data-driven simulation of resilient supplier selection in digital manufacturing. *International Journal of Information Management*, 49, 86–97.

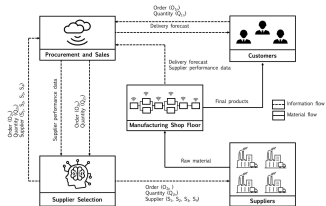


Figure 16: Make to order simulation model

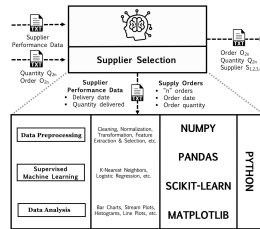


Figure 17: Supplier selection with supervised learning

- One difficulty in managing the resilient supplier portfolios using disruption probability estimations is a relative rarity of risk events which are too intermittent and irregular to be accurately estimated.
- A combination of supervised ML and simulation can help in specifying a risk profile for each order.
- The SC managers can obtain the estimations of what suppliers are most critical in terms of the disruptions and the resulting SC performance impact.

Salmon Aquaculture Case

Prastyabudi, W. A., & Solvang, W. D. (2026). A data-driven predictive modelling of delousing treatment events in Norwegian salmon farming using machine learning. *Results in Engineering*, 30, 110736.

- Sea lice has been one of the major problems of salmon farming, as they negatively impacts both farmed fish and wild salmonid stocks
- Delousing treatments causes negative effects:
 - ▶ the fish growth, leading to smaller size of harvest salmon
 - ▶ contribute to higher mortality rate due to increased fish stress and injury
 - ▶ increase operational costs

Main objective

Develop data-driven ML models for predicting delousing treatment events in the Norwegian salmon aquaculture

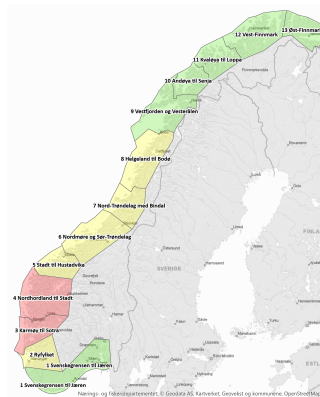


Figure 18: Atlantic salmon aquaculture production areas in Norway.

Machine Learning Models

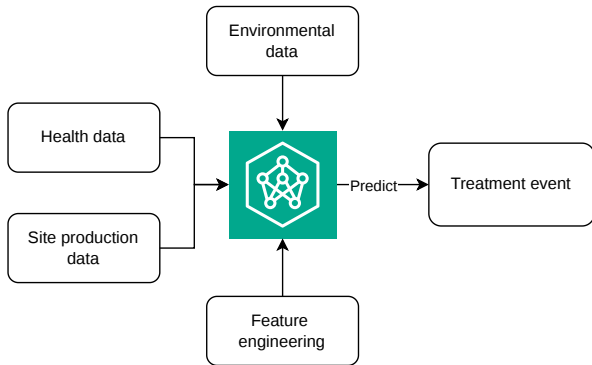


Figure 19: Features used in building ML models.

ML Models

- Logistic Regression (LR)
- Random Forest (RF)
- Adaptive Boosting (AB)
- Gradient Boosting (GB)
- CatBoosting (CB)
- Light Gradient Boosting (LGB)
- Extreme Gradient Boosting (XGB)

- The purpose was to identify a suitable sampling technique for each ML model as the imbalanced datasets significantly hinder the performance of predictive models
- We focused only on the data-driven approach, which performs data resampling through under-sampling, over-sampling, and hybrid sampling.
 - ▶ Cluster-centroid (CC) for under-sampling, which utilizes K-means to assign majority class samples to centroids (Lin et al. 2017),
 - ▶ SMOTE sampling strategy for over-sampling (Chawla et al. 2002)
 - ▶ Hybrid sampling methods: SMOTE + Edited Nearest Neighbours (ENN), SMOTE + Tomek links (Tomek)

Model Training and Evaluation

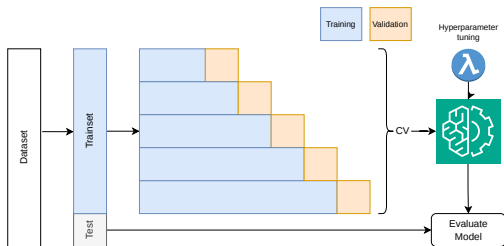


Figure 20: Models training with timeseries cross validation.

- Hyperparameter tuning was implemented through hyperdrive job execution in AzureML.
- We employed random sampling with 50 iterations for the hyperparameter search, using a median stopping policy to prevent overfitting.

Model Evaluation

- Timeseries cross validation (CV) with expanding window was performed
- The efficiency of ML models was assessed using average precision (PR), providing less bias evaluation for imbalanced dataset (Saito and Rehmsmeier 2015).
- AP lift score was computed, which normalized the AP over its corresponding baseline.
- This allows for fair comparison across different baseline.

Experiment Results

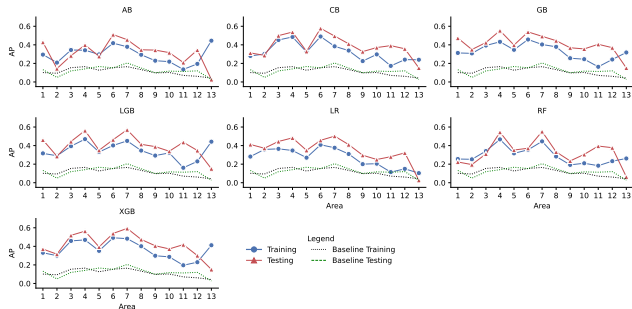


Figure 21: A comparison of model performance on the testing dataset across production areas.

- Ensemble models achieved superior performance compared to the LR model, with the XGB model achieving the highest AP score in 6 out of 13 production areas.

- Sampling strategies improved the models' performance in addressing class imbalance.
- Pairwise comparisons indicate that the AB model was significantly inferior to the GB and LGB models.
- The pairwise comparison revealed that the ensemble models (e.g., GB, LGB, and XGB) had comparable performance.
- Model's performance hinged on sampling strategies, spatial-temporal contexts, and dataset size.

Next Industry AI Case - AI Vision for Quality Inspection

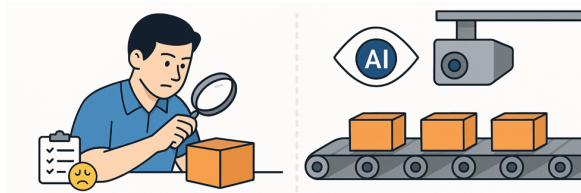


Figure 22: Integrating AI-driven systems to transform quality inspection into smart and automated framework.

Next Industry AI Case - AI Vision for Quality Inspection

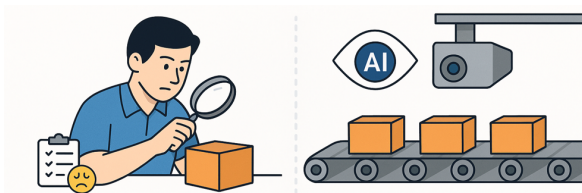


Figure 22: Integrating AI-driven systems to transform quality inspection into smart and automated framework.

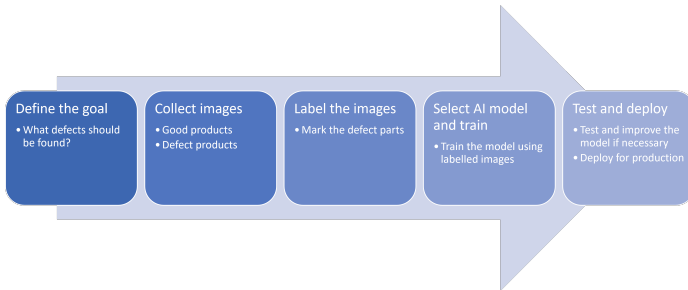


Figure 23: Steps for building object detection using AI.

Existing Platforms for AI Vision

Platform	Strength	Limitation
Roboflow	Easy to use, quick setup, strong community	Less customization for highly specific workflows, offer free tier
V7 (Darwin)	User-friendly, strong annotation tools, good for regulated environments	Smaller ecosystem and community
Supervisely	Flexible workflows, support video and 3D, on-prem options	Higher learning curve, suit for experienced team
Encord	Strong video annotation, good handle of dataset	Cloud-only, pricing with quotation

Final Thoughts

Be Aware

AI does not think like a human. Be aware of its potential risks!

- ML enables systems to learn from data
- Three main types of ML: supervised, unsupervised, reinforcement learning
- Data quality and pre-processing play a critical role in developing good ML models
- Select suitable evaluation metrics based on the problem's requirements
- Significant progress has been made in incorporating ML to solve many industrial and business problems
- AI, including ML, will continue to evolve, affecting daily life and the way we solve problems

Any questions ?

Any questions ?

What can I help with?

Ask anything



Search



Deep research



References

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